

ABSOLUTE OPTICAL POSITION STRIP - INSTALLATION GUIDE

IDENTIFICATION

Machine	GANTRY-01
Project	LINE 4 RETROFIT
Strip ID	AX1-POS-001
Revision	B
Engineer	R. Kelly
Company	Disectron
Generated	2026-07-28 09:13 UTC
Fingerprint	a4f7a81c

Demonstration strip: 421 codes at 25 mm pitch over 10.5 m.

STRIP SPECIFICATION

Symbology	Data Matrix ECC200
Cell pitch	25.000 mm
Symbol size	10.000 mm
Module size	1.0000 mm
Quiet zone	1.000 mm
Margin L/R	7.500 mm
Strip height	40.000 mm
Codes	421
Total length	10565.0 mm (10.565 m)
Max position	10500.0 mm
Per code	25.000 mm
Sheets	39

POSITION FORMULA

Program the PLC with exactly this expression. It is generated from the same geometry that produced the printed strip:

$$P \text{ [mm]} = \text{Index} \times 25.000$$

Each symbol encodes its own absolute position in millimetres, zero padded to 6 digits. The human-readable text below each symbol is the same string the reader decodes.

PRINTING

- Set print scale to EXACTLY 100 % (or 'Actual Size'). This is the most common failure.
- Turn OFF 'Fit to Page', 'Shrink oversized pages' and 'Scale to fit media'.
- Turn OFF any 'Print as image' downsampling; this strip is vector artwork.
- Match the driver resolution to the resolution stated below.
- Print one proof sheet and measure the calibration bar before printing the run.
- Use the same printer, driver and media for every sheet of the run.

Resolution	600 dpi
Scaling	100.000 %
Paper	A4 landscape
Module dots	23.6 (>= 3 required, >= 5 preferred)

CALIBRATION

1. Print page 2 (the first strip sheet) at 100 % scale.
2. Measure the calibration bar end-to-end with a steel rule. Do not use a tape measure.
3. If it measures exactly the nominal length, no correction is needed.
4. Otherwise set Printer Scaling = nominal / measured x 100 % in AOPS and re-export.
5. Re-print and re-measure. The bar must be correct before printing the full run.

THE CALIBRATION BAR MUST MEASURE EXACTLY 200 mm.

MEDIA

- Media: Polyester film, printed by Thermal transfer.
- Predicted substrate movement: 0.42 mm over 10.565 m across a 40 % humidity swing.
- One symbol module spans 23.6 printer dots at 600 dpi (dot size 0.0423 mm). At least 3 dots per module is required; 5 or more is preferred.
- Resin ribbon on synthetic stock is the durable industrial choice: 5+ years with resistance to abrasion, chemicals and UV.
- Clean the mounting surface with isopropyl alcohol and apply from one end, squeegeeing forward to avoid trapped air and longitudinal stretch.

INSTALLATION

1. Clean the mounting surface with isopropyl alcohol and let it dry fully.
2. Establish the machine datum (position 0.000 mm) and mark it clearly on the rail.
3. ALIGN EACH TILE TO ITS PRINTED ABSOLUTE POSITION WITH A STEEL TAPE. Do NOT butt tiles against each other.
4. Butt-splicing would accumulate up to 21.1 mm over 10.565 m; datum alignment bounds the error at 0.54 mm per tile.
5. Each sheet footer prints the absolute X range of that tile - use the leading edge value as the alignment target.
6. Apply from one end, squeegeeing forward, so no air is trapped and the strip is not stretched lengthwise.
7. Keep the strip straight and parallel to the axis of travel within the reader's stated Y tolerance.
8. Leave a deliberate gap at any rail expansion joint; never bridge one with a continuous strip.

READER SELECTION AND MOUNTING

- Required field of view: 35.0 mm (1 x pitch + symbol). Below this there are blind zones in which no complete code is visible and absolute position is lost.
- Static read window (one code at rest): 12.0 mm.
- Module size: 1.000 mm. Sensor: VGA (640 x 480) sufficient (about 175 px across the field of view).
- Working distance by lens: f=6 mm -> 44 mm, f=8 mm -> 58 mm, f=12 mm -> 88 mm, f=16 mm -> 117 mm, f=25 mm -> 182 mm.
- Suggested mount height 86 mm at 12 deg tilt off normal, to avoid specular return.
- Assumes 5.0 pixels per module.
- Assumes a 4.8 mm wide sensor.
- Working distance from the pinhole relation $WD = FOV \times f / \text{sensor width}$.

INSTALLATION GUIDE (CONTINUED 2)

~~Mount height assumes a 12 deg tilt off normal to avoid specular reflection.~~

- These are engineering estimates. Verify against the datasheet of the reader actually selected.

VERIFICATION

- Verify print quality to ISO/IEC 15415; target grade A or B.
- Check that every symbol decodes with the production reader, not a phone.
- Confirm the first and last code on each sheet match the printed code range.
- Confirm the configuration fingerprint on each sheet matches the project file.

WARNINGS

! Do not scale, resample or re-export this PDF in any other application.

! Do not photocopy the strip. Copiers do not hold scale.

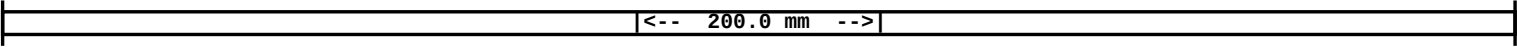
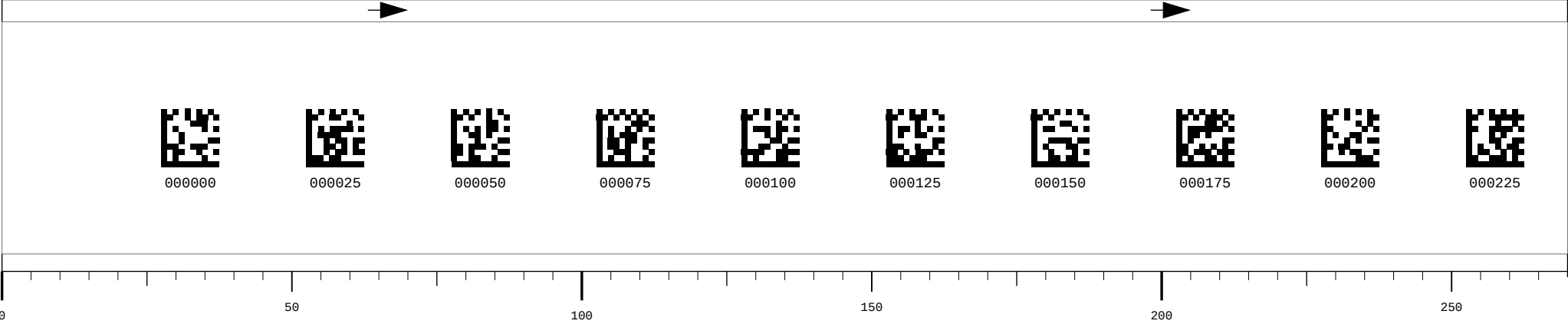
! Cut only in the white gaps. Every page boundary is at least 7.5 mm from any symbol ink (quiet zone required: 1.0 mm).

! The position formula is $P \text{ [mm]} = \text{Index} \times 25.000$. Program the PLC with exactly this expression.

! Verify the first and last positions physically before releasing the axis to automatic operation.

ABSOLUTE OPTICAL POSITION STRIP

MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



THIS BAR MUST MEASURE EXACTLY 200 mm

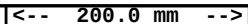
If not: set Printer Scaling = 200 / (measured mm) x 100 %. Printed at 100.000 % scale.



MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



000500



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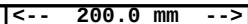




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



000775



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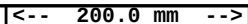




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



001050



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ABSOLUTE OPTICAL POSITION STRIP

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001075


001100


001125


001150


001175


001200

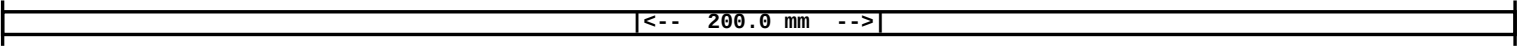

001225


001250


001275


001300


001325



THIS BAR MUST MEASURE EXACTLY 200 mm

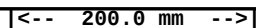
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001600



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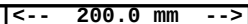




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001875



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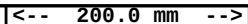




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



002150



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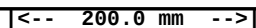




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



002425



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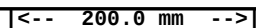




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002700



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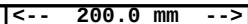




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002975



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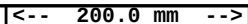




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



003250



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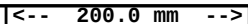




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



003525



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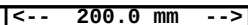




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003800



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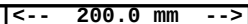




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004075



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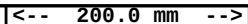




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



004350

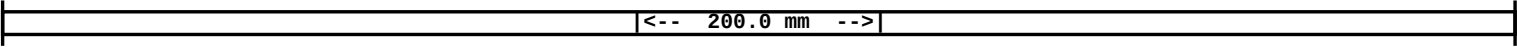
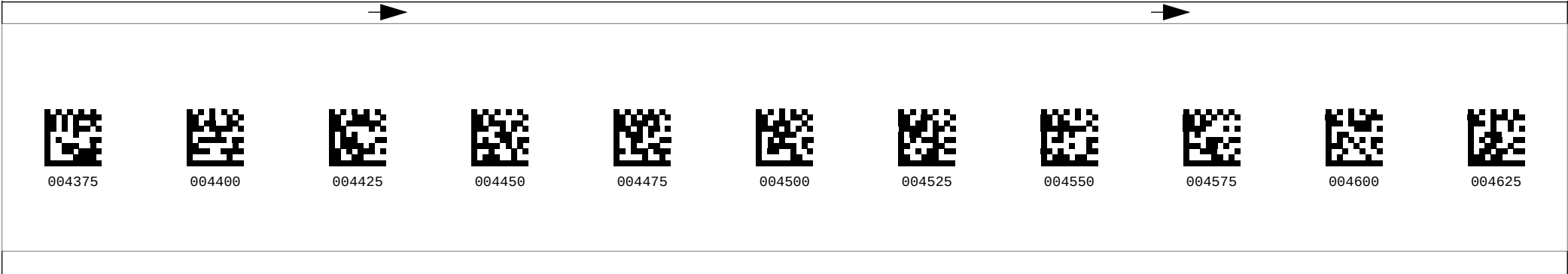


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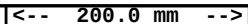
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MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



004900



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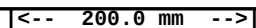




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



005175



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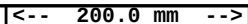




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



005450



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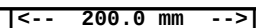




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



005725



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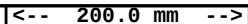




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



006000



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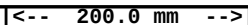




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



006275



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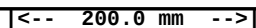




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



006550



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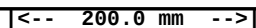




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



006825



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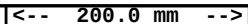




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



007100



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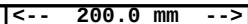




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



007375



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MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



007650



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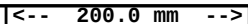




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



007925



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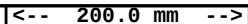




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



008200



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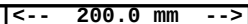




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



008475



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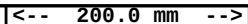




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008750



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009025



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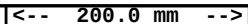




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009300



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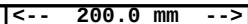




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009575



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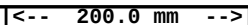




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009850



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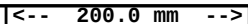




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



010125



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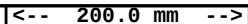




MACHINE GANTRY-01 PROJECT LINE 4 RETROFIT STRIP AX1-POS-001



010400



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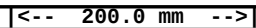




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010500



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